their predecessors. A more correct description is that each new theory refines its predecessor by providing a more accurate and more wide-reaching framework. Newton's theory of gravity has been superseded by Einstein's general relativity, but it would be naïve to say that Newton's theory was wrong. In the domain of objects that don't move anywhere near as fast as light and don't produce gravitational fields anywhere near as strong as those of black holes, Newton's theory is fantastically accurate. Yet this is not to say that Einstein's theory is a minor variant on Newton's; in the course of improving Newton's approach to gravity, Einstein invoked a whole new conceptual schema, one that radically altered our understanding of space and time. But the power of Newton's discovery within the domain he intended it for (planetary motion, commonplace terrestrial motion, and so on) is unassailable.

We envision each new theory taking us closer to the elusive goal of truth, but whether there is an ultimate theory—a theory that cannot be refined further, because it has finally revealed the workings of the universe at the deepest possible level—is a question no one can answer. Even so, the pattern traced out during the last three hundred years of discovery gives tantalizing evidence that such a theory can be developed. Broadly speaking, each new breakthrough has gathered a wider range of physical phenomena under fewer theoretical umbrellas. Newton's discoveries showed that the forces governing planetary motion are the same as those governing the motion of falling objects here on earth. Maxwell's discoveries showed that electricity and magnetism are two sides of the same coin. Einstein's discoveries showed that space and time are as inseparable as Midas' touch and gold. The discoveries of a generation of physicists in the early twentieth century established that myriad mysteries of microphysics could be explained with precision using quantum mechanics. More recently, the discoveries of Glashow, Salam, and Weinberg showed that electromagnetism and the weak nuclear force are two manifestations of a single force—the electroweak force—and there is even tentative, circumstantial evidence that the strong nuclear force may join the electroweak force in a yet grander synthesis.[1] Taking all this together, we see a pattern that goes from complexity to simplicity, a pattern that goes from diversity to unity. The explanatory arrows seem to be converging on a powerful, yet-to-be discovered framework that would unify all of nature's forces and all of